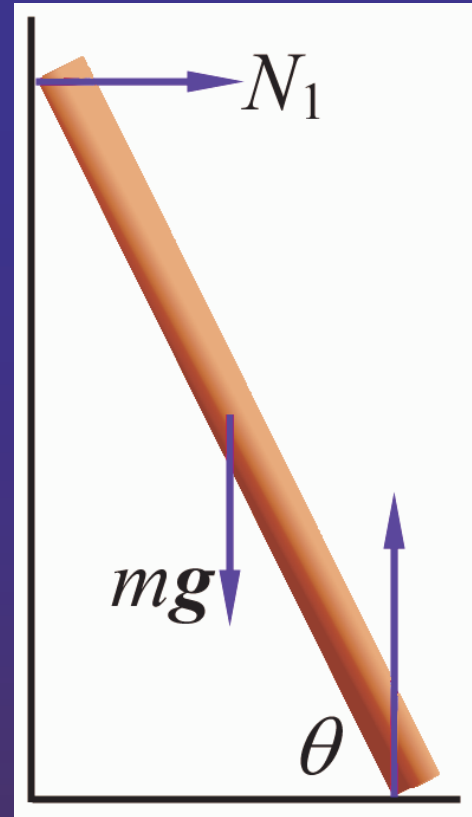


Problem 6.5

Conservation of energy

$$mgL \sin \theta_0 = mgL \sin \theta + \frac{1}{2} mL^2 \dot{\theta}^2 + \frac{1}{2} \left[\frac{1}{12} m(2L)^2 \right] \dot{\theta}^2$$



Motion of com along x : $x = L \cos \theta$,

$$\dot{x} = -L \sin \theta \dot{\theta} \quad \ddot{x} = -L (\cos \theta \dot{\theta}^2 + \sin \theta \ddot{\theta})$$

$$N_1 = -mL (\cos \theta \dot{\theta}^2 + \sin \theta \ddot{\theta})$$

$$N_1 = -mL(\cos \theta \dot{\theta}^2 + \sin \theta \ddot{\theta})$$

Torque equation around 2

$$N_1 \cdot 2L \sin \theta - mgL \cos \theta = \left[\frac{1}{3} m (2L)^2 \right] \ddot{\theta}$$


$N_1 = 0$ leads to

$$\begin{aligned} \dot{\theta}^2 &= -\frac{\sin \theta}{\cos \theta} \ddot{\theta} \\ &= \frac{\sin \theta}{\cos \theta} \frac{mgL \cos \theta}{\frac{4}{3} mL^2} \end{aligned}$$

$$\dot{\theta}^2 = \frac{3g}{4L} \sin \theta$$

Substituting into conservation equation

$$mgL \sin \theta_0 = mgL \sin \theta + \frac{1}{2} mL^2 \dot{\theta}^2 + \frac{1}{2} \left[\frac{1}{12} m(2L)^2 \right] \dot{\theta}^2$$

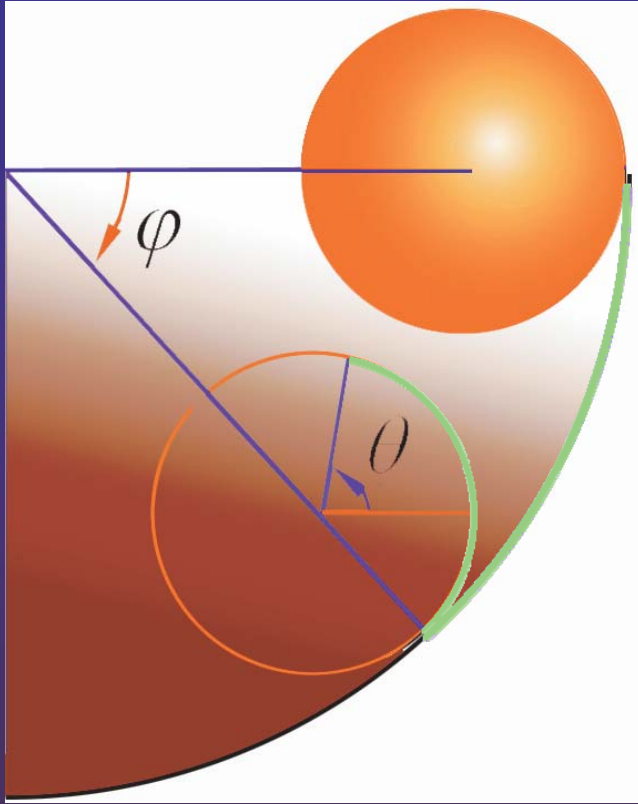
We have

$$\sin \theta_0 = \frac{3}{2} \sin \theta$$

or

$$2L \sin \theta = \frac{2}{3} \times 2L \sin \theta_0$$

Problem 6.6



$$K_C = \frac{1}{2} m [(R - r) \dot{\varphi}]^2$$

$$K_r = ?$$

To find θ ,

$$r(\theta + \varphi) = R\varphi$$

$$\theta = \frac{R}{r} \varphi - \varphi$$

$$\frac{K_r}{K_C} = \frac{\frac{1}{2} \left(\frac{2}{5} m r^2 \right) \left[\left(\frac{R}{r} - 1 \right) \dot{\phi} \right]^2}{\frac{1}{2} m [(R - r) \dot{\phi}]^2} = \frac{2}{5} \dot{\theta}^2$$

Conservation of energy

$$0 = -mg(R - r) + K_C + K_r$$

$$= -mg(R - r) + \frac{7}{5} K_C$$

$$= -mg(R - r) + \frac{7}{2} K_r$$

Equation of motion at φ

$$mg \sin \varphi - N = -m(R - r)\dot{\varphi}^2$$

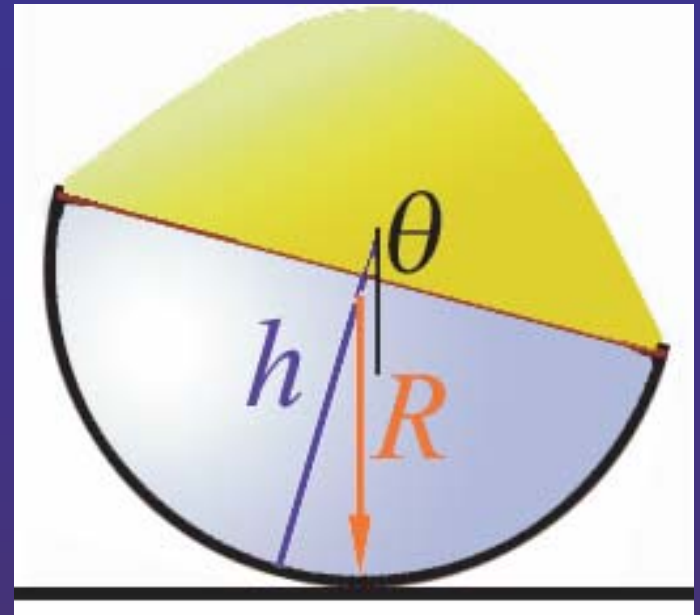
The normal at $\varphi = \pi/2$ is

$$N = \left[mg \sin \varphi + m(R - r)\dot{\varphi}^2 \right] \Big|_{\varphi=\pi/2}$$

$$= mg + \frac{10}{7}mg$$

$$K_C = \frac{1}{2}m[(R - r)\dot{\varphi}]^2 = \frac{5}{7}mg(R - r)$$

Problem 6.12



$$U(\theta) = mg[(h - R)\cos\theta + R]$$

$$\theta = 0, \quad U' = 0, \quad U'' \begin{cases} < 0 & \text{unstable} & h > R \\ = 0 & \text{neutral} & h = R \\ > 0 & \text{stable} & h < R \end{cases}$$

Problem 7.1

In a simple harmonic motion, the equation of motion is:

$$m\ddot{x} + kx = 0 \quad \text{or} \quad m\ddot{x} = -kx$$

Kinetic energy and potential energy are respectively

$$\frac{1}{2}m\dot{x}^2$$

$$\frac{1}{2}kx^2$$

(a)

$$\frac{1}{T} = \frac{\omega_0}{2\pi}$$

$$\frac{1}{T} \int_0^T \frac{1}{2} kx^2 dt = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} kA^2 \cos^2(\omega_0 t + \varphi) d(\omega_0 t)$$

$$= \frac{1}{4} kA^2$$

$$\frac{1}{T} \int_0^T \frac{1}{2} m\dot{x}^2 dt = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} mA^2 \omega_0^2 \sin^2(\omega_0 t + \varphi) d(\omega_0 t)$$

$$= \frac{1}{4} kA^2$$

(b)

$$\frac{1}{A} \int_0^A \frac{1}{2} kx^2 dx = \frac{1}{6} kA^2$$

$$\frac{1}{A} \int_0^A \frac{1}{2} m\dot{x}^2 dx = \frac{1}{\omega_0 A} \int_0^{\pi/2} \frac{1}{2} m\dot{x}^3 d(\omega_0 t)$$

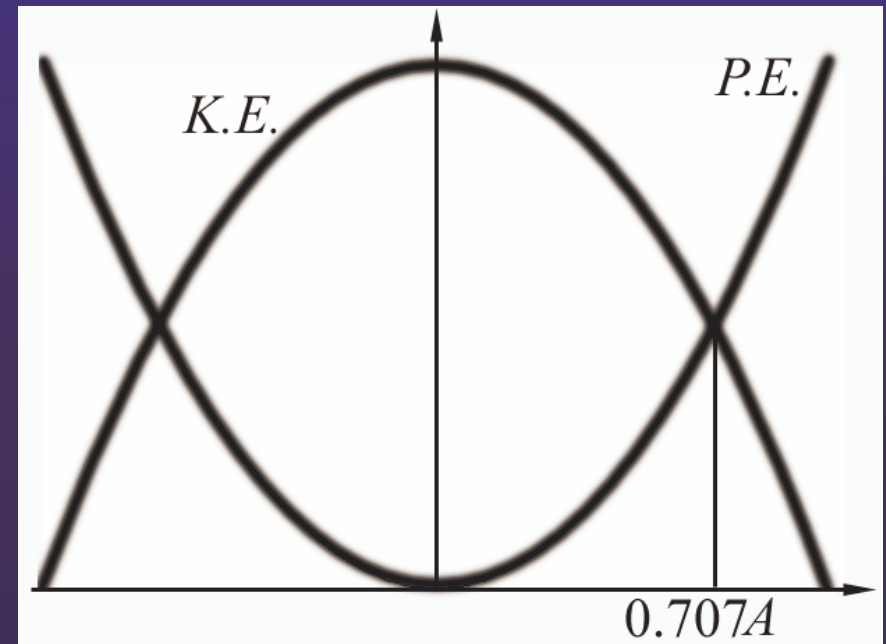
$$= \frac{1}{\omega_0 A} \frac{1}{2} m\omega_0^3 A^3 \int_0^{\pi/2} \sin^3(\omega_0 t + \varphi) d(\omega_0 t)$$

$$= \frac{1}{3} kA^2$$

(c)

$PE \leq KE$ leads to $|x| \leq 0.707A$, while in (a) larger part of PE with a larger weight

$$dt = \frac{1}{\dot{x}} dx$$



Problem 7.3

(see problem 6.6 for general solution)

$$-mg \sin \theta - F_s = mR\ddot{\theta}$$

$$bF_s = \frac{2}{5}mb^2\ddot{\phi}$$

For $b \ll R$, the geometric relation is similar to rolling ball on the planar surface:

$$R\theta \approx b\phi$$

$$-mg \sin \theta - \frac{2}{5}mR\ddot{\theta} = mR\ddot{\theta}$$

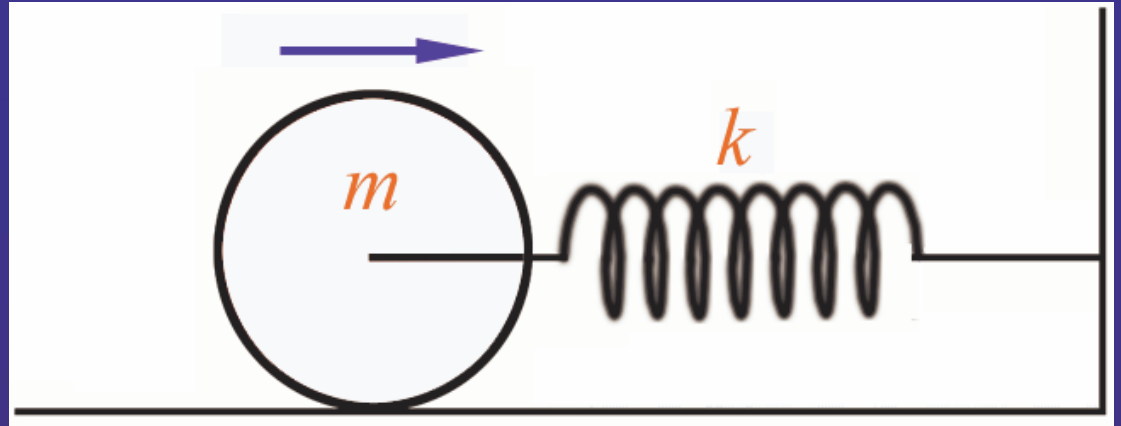
$$-mg \sin \theta = \frac{7}{5}mR\ddot{\theta}$$

$$-g \sin \theta = \frac{7}{5}R\ddot{\theta}$$

$$\ddot{\theta} + \frac{5g}{7R}\theta = 0$$

$$\omega = \sqrt{\frac{5g}{7R}}$$

Problem 7.4



$$\frac{1}{2}kA^2 = K_T + K_r + \frac{1}{2}k0^2$$

$$K_r = \frac{1}{2} \cdot \frac{1}{2}mr^2\dot{\theta}^2 = \frac{1}{4}m\dot{x}^2 = \frac{1}{2}K_T$$

$$K_T = \frac{1}{3}kA^2 = \frac{1}{16}J$$

$$m\ddot{x} = -kx + f$$

$$\frac{1}{2}mr^2\ddot{\theta} = -fr$$

$$\frac{1}{2}mr\ddot{\theta} = -f$$

$$rd\theta = dx$$

$$\frac{3}{2}m\ddot{x} = -kx$$

$$\ddot{x} + \frac{2k}{3m}x = 0$$

$$T = 2\pi\sqrt{\frac{3m}{2k}}$$

***Problem 7.5**

(See the reference cited) When the spring is stretched to a length X , the kinetic energy of the *spring* alone is

$$\frac{1}{2} \int_0^X \left(\frac{m_s}{X} dx \right) \dot{x}^2 = \frac{1}{2} \frac{m_s}{X} \int_0^X dx \left(\frac{x}{X} \dot{X} \right)^2 = \frac{1}{6} m_s \dot{X}^2$$

$$X = N \Delta \ell; x = n \Delta \ell \text{ (uniformly stretched)}$$

$$\dot{x} = \frac{d}{dt}(n \Delta \ell) = \frac{d}{dt} \left(n \frac{X}{N} \right) = \frac{n}{N} \dot{X} = \frac{x}{X} \dot{X}$$

The conservation of energy reads

$$-mg(X - \ell) - m_s g \left(\frac{X}{2} - \frac{\ell}{2} \right) + \frac{1}{2} k (X - \ell)^2 \\ + \frac{1}{2} m \dot{X}^2 + \frac{1}{6} m_s \dot{X}^2 = 0$$

$$\frac{d}{dt}$$

$$\left(m + \frac{1}{3} m_s \right) \ddot{X} = -k(X - \ell) \dot{X} + \left(m + \frac{1}{2} m_s \right) g \dot{X}$$

$\dot{X} = 0$ is the trivial solution. Then

$$\left(m + \frac{1}{3}m_s\right)\ddot{X} = -k(X - \ell) + \left(m + \frac{1}{2}m_s\right)g$$

Problem 7.7

$$Ae^{-\gamma t} = \frac{1}{3}A$$

$$t = \ln 3 / \gamma \quad \gamma = \frac{b}{2m}$$

$$n = \frac{\omega}{2\pi} t = \frac{\ln 3}{2\pi} \sqrt{\left(\frac{\omega_0}{\gamma}\right)^2 - 1}$$

$$= \frac{\ln 3}{2\pi} \sqrt{\frac{4mk}{b^2} - 1}$$

$$= 5.264 \longrightarrow 6$$

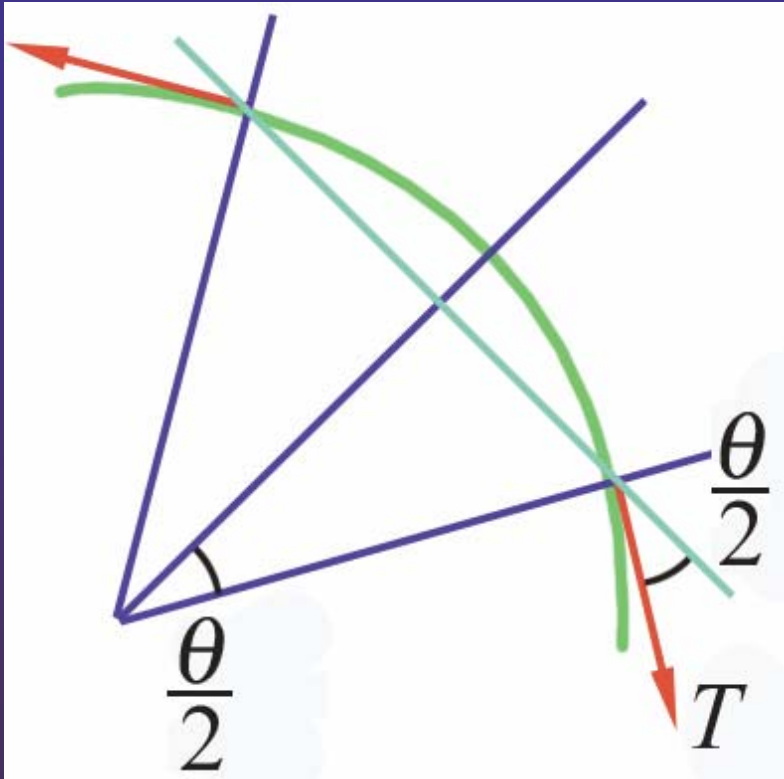
Problem 7.8

$$\int_0^T \dot{x} dt = \int_{\omega t=0}^{2\pi} dx \rightarrow m\ddot{x} + kx = F + F_\gamma$$

$$0 = \frac{1}{2} m \dot{x}^2 \Big|_{\omega t=0}^{2\pi} + \frac{1}{2} k x^2 \Big|_{\omega t=0}^{2\pi} = \int F \dot{x} dt + \int F_\gamma \dot{x} dt$$

Problem 8.1

Consider a short arc, equation of motion reads



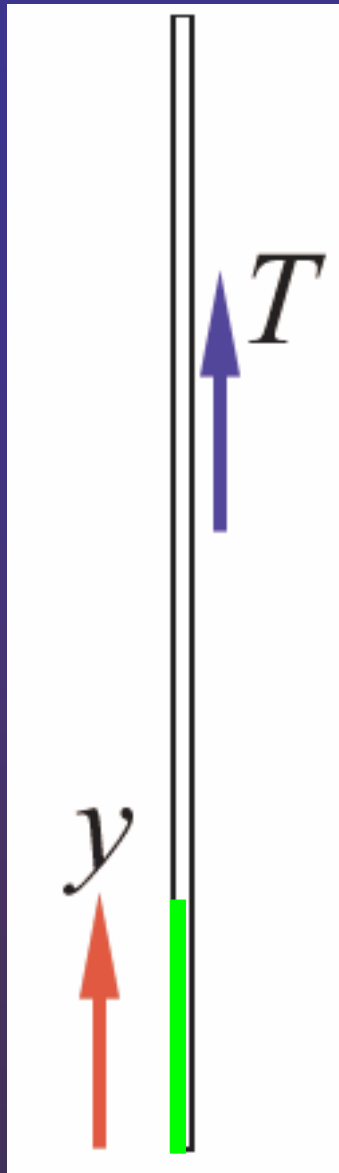
$$v = \sqrt{\frac{T}{\rho}}$$

$$2T \sin \frac{\theta}{2} = (\rho R \theta) \frac{v_0^2}{R}$$

$$\sin \frac{\theta}{2} \approx \frac{\theta}{2}, \quad T = \rho v_0^2,$$

$$v = \sqrt{\frac{T}{\rho}} = v_0$$

Problem 8.2



$$T(y) = \frac{m}{L} yg$$

$$v = \sqrt{\frac{T}{m/L}} = \sqrt{gy}$$

$$t = \int dt = \int_0^y \frac{dy}{v} = 2\sqrt{L/g}$$

Problem 8.14

(a) node

$$(b) \quad R = \frac{\lambda}{4} = \frac{vT}{4} \qquad T = \frac{4R}{v}$$

$$(c) \quad v = \sqrt{\gamma \frac{P}{\rho}} = 1.15 \times 10^6 \text{ m/s}$$

$$T = 22\text{s}$$

Problem 8.15 Known $v = 80\text{cm/s}$, $x = 10\text{cm}$, ρ

$$u = 5.0 \sin(1.0 - 4.0t) = 5.0 \cos\left(4.0t - 1.0 + \frac{\pi}{2}\right)$$

$$4.0t - 1.0 + \frac{\pi}{2} = \omega t - k \cdot 10 + \varphi; \quad \omega = 4.0$$

$$\text{So } k = \frac{\omega}{v} = \frac{4.0}{80} = 0.05 \quad \varphi = \frac{\pi}{2} - 0.5$$

$$u = 5.0 \cos\left(4.0t - 0.05x + \frac{\pi}{2} - 0.5\right)$$

$$4.0t - 1.0 + \frac{\pi}{2} = \omega t + k \cdot 10 + \varphi; \quad \omega = 4.0$$

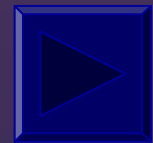
So $k = \frac{\omega}{v} = \frac{4.0}{80} = 0.05 \quad \varphi = \frac{\pi}{2} - 1.5$

$$u = 5.0 \cos \left(4.0t + 0.05x + \frac{\pi}{2} - 1.5 \right)$$

$$T = v^2 \rho$$

$$= (80 \text{ cm/s})^2 \times 4.0 \text{ g/cm}$$

$$= 64 \times 4.0 \times 10^{-3} \text{ N} = 0.26 \text{ N}$$



Problem 8.17

$$u_1 = \frac{A}{r_1} \cos(kr_1 - \omega t)$$

$$u_2 = \frac{A}{r_2} \cos(kr_2 - \omega t)$$

where

$\frac{A}{r}$ is the amplitude of a spherical wave

For $r_1 \approx r_2$

we may define $r \equiv \frac{r_1 + r_2}{2}$ $\Delta r \equiv r_1 - r_2$

So

$$r_1 = r + \frac{\Delta r}{2}$$

$$r_2 = r - \frac{\Delta r}{2}$$

The superposition is

$$u_1 + u_2 = \frac{A}{r + \frac{\Delta r}{2}} \cos\left(kr - \omega t + \frac{k\Delta r}{2}\right) \\ + \frac{A}{r - \frac{\Delta r}{2}} \cos\left(kr - \omega t - \frac{k\Delta r}{2}\right)$$

$$\begin{aligned} &= \left(\frac{A}{r + \frac{\Delta r}{2}} + \frac{A}{r - \frac{\Delta r}{2}} \right) \cos\left(\frac{k\Delta r}{2}\right) \cos(kr - \omega t) \\ &\quad - \left(\frac{A}{r + \frac{\Delta r}{2}} - \frac{A}{r - \frac{\Delta r}{2}} \right) \sin\left(\frac{k\Delta r}{2}\right) \sin(kr - \omega t) \\ &\approx \left(\frac{2A}{r} \right) \cos\left(\frac{k\Delta r}{2}\right) \cos(kr - \omega t) \end{aligned}$$

Problem 8.21

$$\sin \frac{120}{2} = \frac{\frac{3}{4}c}{u}$$

$$u = \frac{\sqrt{3}}{2}c = 2.6 \times 10^8 \text{ m/s}$$

Problem 8.22 Microwave is EM wave.
Optical Doppler formula should be used.
The incident microwave is blueshifted due to
approaching (receiver) airplane

$$\nu' = \sqrt{\frac{1 + \frac{u}{c}}{1 - \frac{u}{c}}} \nu$$

**while reflecting microwave is also blueshifted
due to approaching (source) airplane**

$$\nu'' = \sqrt{\frac{1 + \frac{u}{c}}{1 - \frac{u}{c}}} \nu' = \frac{1 + \frac{u}{c}}{1 - \frac{u}{c}} \nu \approx \nu \left(1 + \frac{2u}{c} + 2 \left(\frac{u}{c} \right)^2 + \dots \right)$$

On the other hand

$$\nu_B = \frac{\nu'' - \nu}{2} \times 2$$

$$\nu'' = \nu \left(1 + \frac{2u}{c} \right) = \nu + \nu_B = \nu \left(1 + \frac{\nu_B}{\nu} \right)$$

Thus

$$u = \frac{\nu_B}{2\nu} c = \frac{\nu_B}{2} \lambda$$

$$= \frac{990 \text{ Hz}}{2} \times 0.10 \text{ m} = 49.5 \text{ m/s}$$